

A V-Band Four-Channel Phased Array Transmitter Front-End With 0.7° Phase Step and 20 dB Gain Dynamic Range

Cheol So^{ID}, *Member, IEEE*, Eun-Taek Sung^{ID}, *Member, IEEE*, and Songcheol Hong^{ID}, *Member, IEEE*

Abstract—This article reports a V-band phased array transmitter front-end module, in which a four-channel RF beamforming IC is included with power amplifiers (PAs), active phase shifters, and power dividers. The PA is linearized by a cold-FET-controlled coupled inductor, improving AM to AM and AM to PM properties. The vector sum of I and Q signals from an active I/Q generator determines the amplitude and phase of a synthesized signal. The non-ideal effects of the vector summing circuit are overcome by introducing a 45° phase shifter implemented with a C-L-C network. It shows 0.1 dB rms gain and 2° rms phase errors at 60 GHz, with 3-bit gain and 9-bit phase controls in the 20 dB dynamic range. It achieves an OP1dB of 10.8 dBm and a peak gain of 14.3 dB in the 3-dB frequency bandwidth of 57–63.3 GHz. A 10 mil thick antenna substrate is used to implement four-channel series-fed patch antennas. The four-channel chip is connected to the antenna module by wire bonding, of which effects are compensated by impedance matching networks on the substrate. The entire antenna feeding network shows less than 2 dB insertion loss, which shows loss and phase variations of 0.3 dB and 3.5° at 60 GHz, respectively. The far-field beam pattern of the four-channel transmitter module is also measured.

Index Terms—Active R-L poly-phase filter, C-L-C network, chip to antenna interconnection, CMOS, cold-FET, coupled inductor, linearizer, millimeter-wave (mm-wave), phased-array, transmitter.

I. INTRODUCTION

THE 60 GHz frequency band is an unlicensed band that offers high data rates in communication [1], [2], [3], [4] and high-range resolution in radar [5], [6], [7], [8], [9]. However, the frequency band suffers from high atmospheric attenuation and path loss. To overcome these challenges, beamforming techniques are actively introduced to improve

Manuscript received 23 September 2023; revised 21 December 2023 and 12 February 2024; accepted 13 February 2024. Date of publication 18 March 2024; date of current version 7 May 2024. This work was supported in part by the Institute of Information and Communications Technology Planning and Evaluation (IITP) funded by the Korean Government, Ministry of Science and ICT (MSIT) under Grant 2019-0-00826; and in part by Samsung Electronics Company Ltd. This article is an expanded version from the IEEE RFIC Symposium, San Diego, CA, USA, June 11–13, 2023 [DOI: 10.1109/RFIC54547.2023.10186121]. (Corresponding author: Cheol So.)

Cheol So is with the Department of Electrical and Computer Engineering, University of California at Santa Barbara, Santa Barbara, CA 93106 USA (e-mail: cheolso93@gmail.com).

Eun-Taek Sung and Songcheol Hong are with the Department of Electrical Engineering, Korea Advanced Institute of Science and Technology, Daejeon 34141, South Korea (e-mail: schong@ee.kaist.ac.kr).

Color versions of one or more figures in this article are available at <https://doi.org/10.1109/TMTT.2024.3373752>.

Digital Object Identifier 10.1109/TMTT.2024.3373752

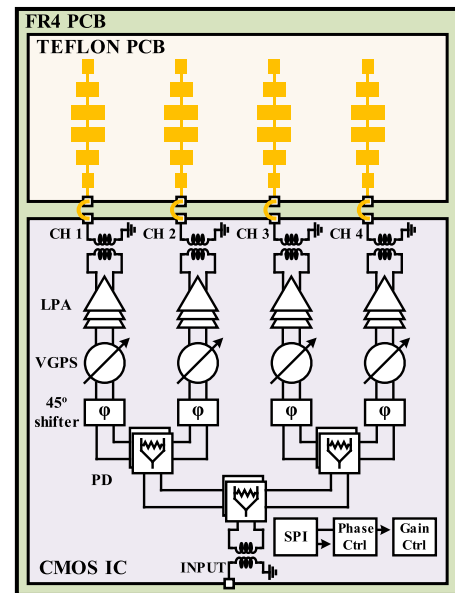


Fig. 1. Block diagram of the proposed V-band four-channel transmitter beamforming module.

the signal-to-noise ratio (SNR) by aligning the beams from a phased antenna array in specific directions [10], [11], [12], [13], [14], [15], [16]. As the number of channels increases N times, the total power consumption of the transmitter also increases, while the EIRP increases by N^2 times. This results in an overall improvement in TX efficiency by N times [17]. The improved angular resolution of a system allows it to be robust against interference signals. In addition, as the output power required from each channel is reduced, it is possible to implement the beamforming front-end with a CMOS chip, resulting in small and cost-effective systems.

However, as the number of array elements increases, not only does the EIRP of the mainlobe increase, but the EIRP of the sidelobe also increases. As a result, the interference through the sidelobe is not negligible. Especially, for implementing a high rejection ratio spatial filter that rejects interference and allows desired signals from specific angle directions, precise control of phase and gain at each channel is required [18]. Therefore, high phase resolution is necessary to keep the sidelobe level low and reduce interference signals from undesired directions. Since amplitude shaping in each array can lower further the sidelobe level, an additional gain control function is required [19], [20], [21].

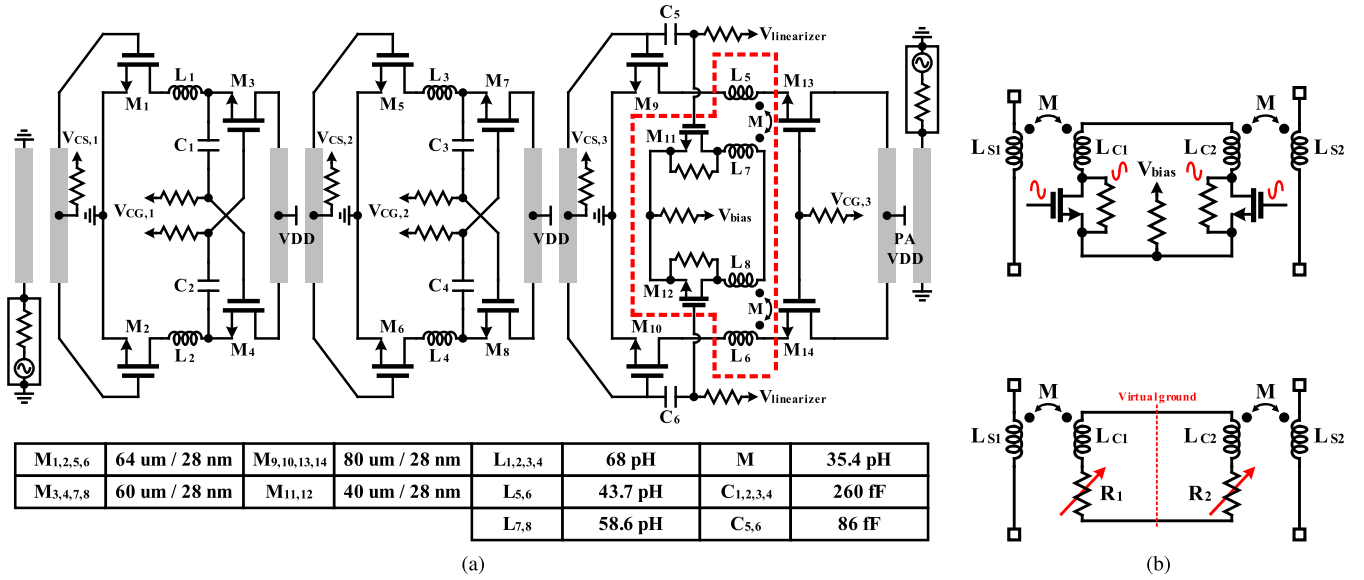


Fig. 2. Proposed linear PA. (a) Schematics and (b) cold-FET-controlled coupled inductor linearizer at inter-stage node, and its equivalent circuits.

The descriptions of the power amplifier (PA) and phase shifter in the V-band four-channel transmitter beamforming IC were reported in [22]. This article presents more detailed descriptions and analyses of the building blocks in the beamforming IC and the properties of the transmitter beamforming module based on the chip, including the chip to antenna interconnection. A cascode PA is proposed to improve the AM to AM and AM to PM performances, in which cold-FET-controlled variable inductors are introduced in the inter-stage nodes for linearization. In the phase shifter, an active vector modulator structure is adopted [23], [24], [25], [26], [27]. The output signal is synthesized by summing the I and Q vectors, of which amplitudes are determined by the DACs. The non-idealities in the vector modulator-based phase shifter are addressed, and C-L-C network-based 45° phase shifter is proposed to mitigate this problem.

A six element series-fed patch antenna array is fabricated on a Teflon substrate and connected to the chip through bond wires and their compensation structures on the substrate to verify the performance. To minimize the length of the bond wire, the heights of the chip and the antenna substrate are designed to be matched [28]. In addition, bond wires are measured to be modeled, of which effects are compensated by introducing an additional impedance transforming network on the substrate. Finally, a four-channel transmitter beamforming module is fabricated, and its far-field beam pattern is measured.

Section II presents the four-channel transmitter beamforming IC structure. Section III provides the transmitter beamforming module. Section IV reports the implementations and measurements of the IC and the module, including far-field beam patterns and EIRP. Section V draws conclusions.

II. TRANSMITTER BEAMFORMING IC STRUCTURE AND ITS RF BUILDING BLOCKS

Fig. 1 shows a block diagram of the proposed V-band four-channel beamforming transmitter module. The signal is

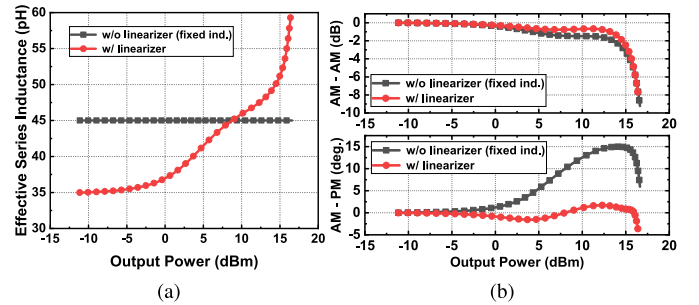


Fig. 3. Simulated results of the inter-stage inductor linearizer. (a) Effective inter-stage series inductance. (b) AM to AM and AM to PM of the PA with the linearizer on and off.

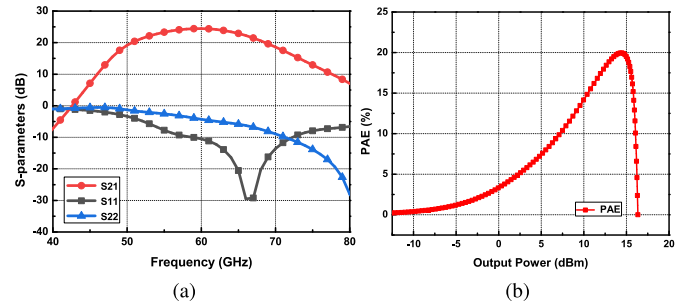


Fig. 4. Simulated (a) S-parameters and (b) PAE of the linear PA.

applied to the beamforming IC and distributed to the four channels via 1:2 differential power dividers. Each channel consists of a phase shifter and PA. The phase shifter determines the specific gain and the phase state. The amplified signals are applied to the series-fed patch antennas on the Teflon substrate via bond wires.

A. PA Linearized by Cold-FET-Controlled Coupled Inductor

Since the scaling of advanced CMOS transistors leads to decreases in their breakdown voltages, achieving a high-output

power from a CMOS PA becomes difficult due to its low-supply voltage. Thus, the stacked CMOS transistor structures are frequently used to design a PA for high-output power. However, the gain and efficiency are reduced due to parasitic capacitances in the internode of the stacked structure. Series or parallel inductors are introduced to the internodes to cancel out the parasitic capacitances and improve the gain and efficiency [29], [30].

Also, millimeter-wave (mm-wave) CMOS PA operates at a class-AB bias to achieve high efficiency and linearity [31], [32], [33]. This causes a gain compression with many harmonics at high powers. In addition, as the input power increases, the average gate-source capacitance of a CS amplifier decreases, which leads to a phase distortion of a PA [34]. Here, a cold-FET-controlled coupled inductor linearization technique is proposed to address the gain compression and phase distortion of a PA.

Fig. 2(a) shows the schematic of the proposed linear PA. It consists of two-stage driver amplifier and a power stage amplifier. To obtain a sufficient gain, the cross-coupled capacitor technique is applied to the cascode driver amplifiers to boost the gm of the common gate stage [35]. Fig. 2(b) shows the proposed cold-FET-controlled coupled inductor linearizer and its equivalent circuits. It consists of the series inductance (L_S), coupled inductance (L_C), and a cold-FET transistor which is connected to L_C . The effective resistance of the cold-FET transistor increases as the amplitude of the swing at the drain node increases when the same dc voltage is applied to the drain and source nodes with a specific gate bias voltage [36], [37]. In the proposed structure, the voltage swing of the opposite phase is applied to drain and gate nodes, resulting in more changes in effective resistance when the same voltage swing is applied. By expressing the cold-FET as a variable resistance R , the effective inter-stage series inductance $L_{S,eff}$ can be expressed as

$$L_{S,eff} = L_S - \frac{w^2 M^2 L_C}{R^2 + w^2 L_C^2}. \quad (1)$$

Fig. 3 shows the simulated results of the inter-stage inductor linearizer. As input power increases, R also increases, leading to a larger inductance L_S . It helps to cancel out the parasitic capacitances at the internode, resulting in improved AM to AM properties. Additionally, the increase of series inductance causes a phase lag which compensates the AM to PM distortion of the PA. Fig. 4 shows the simulated S-parameters and PAE of the linear PA. It achieves a peak gain of 24.4 dB.

B. Variable Gain Active Phase Shifter

Fig. 5 shows the schematic of the variable gain active phase shifter. The active I/Q generator is used, which enables a finite gain in the I/Q generator by placing an R-L poly-phase filter in the inter-stage node of a cascode amplifier without an additional impedance matching network [27]. The I and Q vectors are fed to the vector summing amplifier, generating I positive, I negative, Q positive, and Q negative vectors. The amplitude of each vector is determined by the gate bias of the current

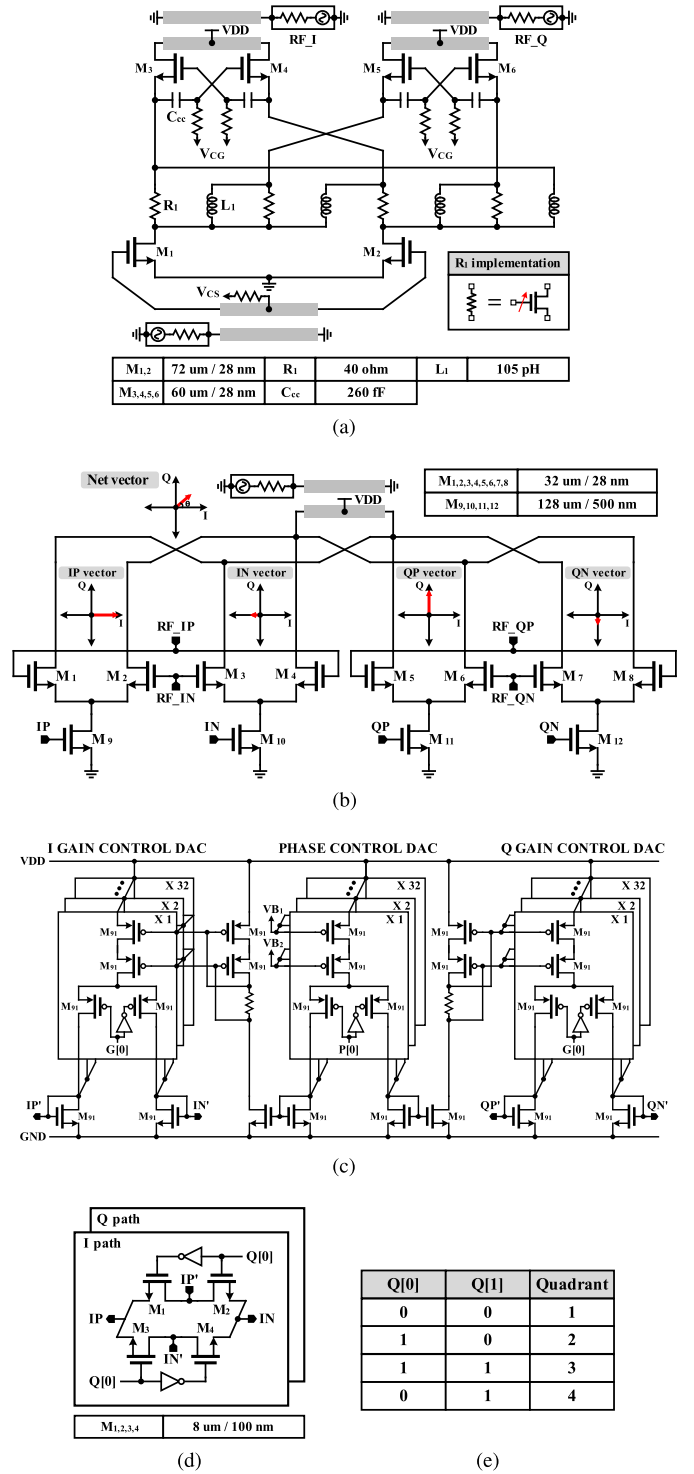


Fig. 5. Schematics of the vector modulator. (a) Active I/Q generator, (b) vector summing amplifier, (c) current steering DACs for phase and gain selections, (d) DPDT switches for quadrant selection, and (e) quadrant selection logic.

source, which is controlled by phase and gain control DACs. The DAC unit has a binary weighted 6-bit current steering DAC structure. To make the DAC linear, the unit length of the transistor is designed to be 500 nm. It flows the same total dc current regardless of all gain and phase states through complementary operations. The phase control DAC determines the ratio of I and Q vectors, and the gain control DAC

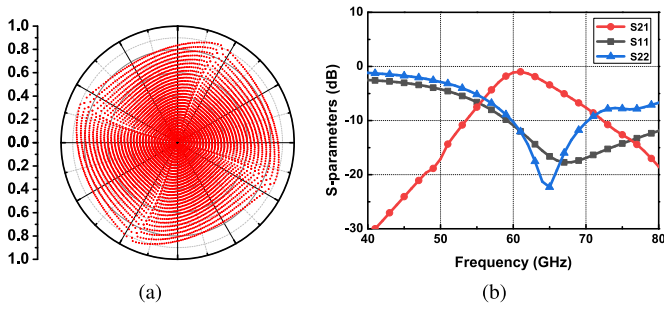


Fig. 6. Simulated results of the active phase shifter. (a) Constellation at 60 GHz and (b) S-parameters at maximum gain state.

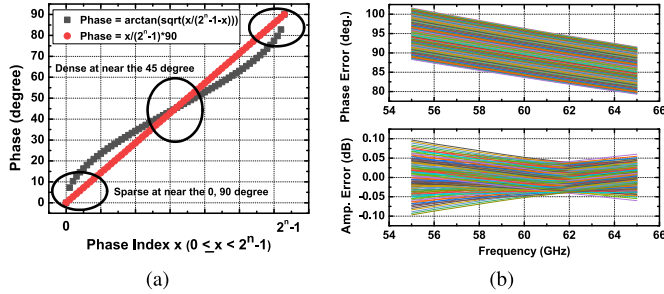


Fig. 7. Non-ideal effects of the vector sum phase shifter. (a) Arc tangent nonlinearity of phase and (b) phase and amplitude errors due to load impedance variations of I/Q generator with respect to gain and phase states.

vector determines the ratio of positive and negative vectors for each I and Q. The effective transconductance and phase are expressed as

$$G_{\text{meff}} = \sqrt{2\mu_{\text{eff}}C_{\text{ox}} \frac{W}{L} I_{\text{LSB}}} \times \left(\sqrt{\sum_{g=0}^5 2^g \times G[g]} - \sqrt{\sum_{g=0}^5 2^g \times \overline{G[g]}} \right) \quad (2)$$

$$\text{Phase} = \arctan \left(\frac{\sqrt{\sum_{p=0}^5 2^p \times \overline{P[p]}}}{\sqrt{\sum_{p=0}^5 2^p \times P[p]}} \right). \quad (3)$$

$G[n]$ and $P[n]$ represent the n th bit in the binary-weighted gain control DAC and phase control DAC, respectively, which have 0 or 1. By changing the positive and negative vector polarity using the DPDT switches between the DAC and the vector summing amplifier, the gain and phase of the other three quadrants can also be configured [23], [24], [27]. Fig. 6 shows the simulation results of the variable gain active phase shifter. It achieves 8-bit phase resolution in 360° and a peak gain of -1 dB.

Fig. 7 shows the non-ideal effects of the vector sum phase shifter. The arc tangent function of the ratio of the I and Q vectors determines the phase. As a result, the phase step increases near 0° and 90° . For example, when a 6-bit resolution represents a quadrant, the phase step is 1.4° in a linear step, while the maximum phase step is 7.2° in the vector summing phase shifter.

In addition, the current flowing through the current source changes depending on the gain and phase state in the vector

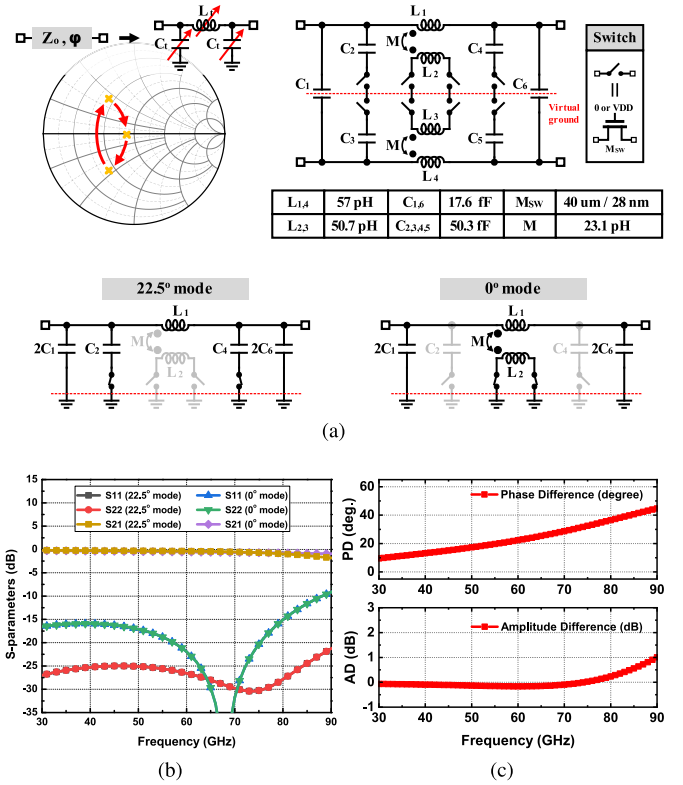


Fig. 8. Two-stage C-L-C 45° phase shifter. (a) Schematic of the C-L-C network based 22.5° phase shifter, (b) simulated S-parameters, and (c) amplitude and phase difference of the 22.5° phase shifter.

summing amplifier. In Fig. 5(b), the input impedance changes as the operating region of M_{1-8} changes from the saturation to the triode region, which causes a change in the load impedances of the I/Q generator, resulting in I/Q errors. When the real and imaginary parts of the load impedance change by 20%, the amplitude and phase errors of the I/Q generator can range from 0.12 dB and 12.8° , respectively. It means that the I and Q axes change depending on the gain and phase state in the constellation. Therefore, there is a limitation in expressing high phase resolution using a vector summing active phase shifter.

C. 45° Phase Shifter for High Phase Resolution

Due to the abovementioned phase nonlinearity of the vector modulator, the constellations near 0° and 90° show large gain and phase errors. Additionally, I/Q errors due to input impedance variations of the vector summing amplifier also cause large gain and phase errors near 0° and 90° . To address the limitations of the vector summing phase shifter, a 45° phase shifter is proposed. By shifting the 45° in the constellation, the phase steps near 0° and 90° are expressed densely. A switched transmission line-based phase shifter is introduced to minimize phase mismatch between two phase modes due to the return loss differences. It is implemented as a π -type C-L-C artificial transmission line for a compact size [38], [39]. Fig. 8 shows the schematic and simulation results of the implemented C-L-C network based 22.5° phase shifter. Due

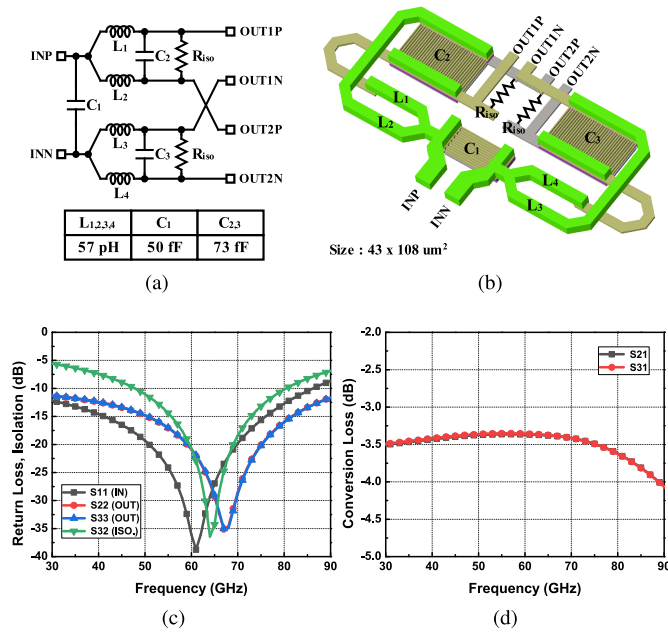


Fig. 9. 1:2 differential power divider in the front end of the beamforming IC. (a) Schematic, (b) layout, (c) simulated return loss and isolation, and (d) conversion loss.

to the cut-off frequency of the C-L-C network, the 45° phase shifter is implemented using two stages of 22.5° phase shifters. To have the phase shift of an angle ϕ , the capacitance (C_t) and inductance (L_t) in the C-L-C network can be expressed as

$$C_t = \frac{1 - \cos(\phi)}{w Z_o \sin(\phi)}, \quad L_t = \frac{Z_o \sin(\phi)}{w}. \quad (4)$$

To implement an effective transmission line with two different phase modes, switched capacitor and switched coupled inductor are used. C and L in each phase mode can be expressed as

$$C_{t,22.5^\circ} = 2C_1 + C_2, \quad L_{t,22.5^\circ} = L_1 \quad (5)$$

$$C_{t,0^\circ} = 2C_1, \quad L_{t,0^\circ} = L_1 - \frac{M^2}{L_2}. \quad (6)$$

C_1 , C_2 , L_1 , L_2 , and M are set to make ϕ_1 - ϕ_2 to be 22.5° . The return loss of the 22.5° phase shifter is less than -15 dB, while the phase and amplitude differences are 22.3° and 0.16 dB at 60 GHz, respectively.

D. 1:2 Differential Power Divider

Fig. 9 shows the proposed 1:2 differential power divider. It is designed with lumped capacitors and inductors instead of a $\lambda/4$ transmission line, resulting in a compact size. As a fully differential structure, it operates insensitively with ground conditions. By designing with a differential 50Ω (25Ω in each single-ended path), the size of the inductor can be reduced [40]. Fig. 9(c) and (d) shows the simulated S-parameters of the 1:2 power divider. Input and output return losses and the isolation achieve less than -15 dB at 60 GHz. The conversion loss, excluding the dividing loss of 3 dB, is an additional 0.4 dB.

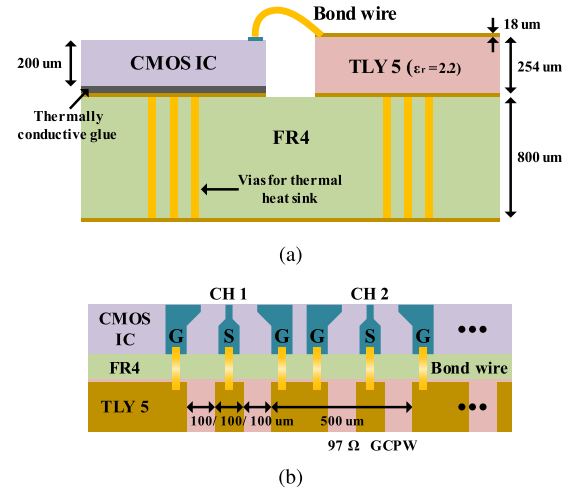


Fig. 10. Chip to antenna interconnection. (a) Cross section view and (b) top view.

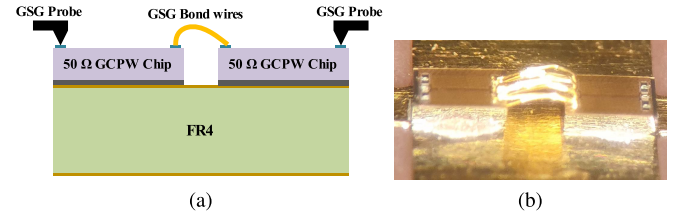


Fig. 11. Test pattern to model the GSG bond wires. (a) Cross section view and (b) its microphotograph.

III. TRANSMITTER BEAMFORMING MODULE

Fig. 10(a) shows the cross section view of the chip to antenna interconnection. The length of the bond wire is minimized by designing the antennas on a Teflon substrate of similar height to the beamforming IC. The beamforming IC, with a height of $200 \mu\text{m}$, is attached to the FR4 substrate using thermally conductive glue. The antenna substrate of 10 mil ($254 \mu\text{m}$) thickness is selected considering the height of the beamforming IC. Fig. 10(b) shows the top view of the chip to antenna interconnection. The minimum resolution of the low-cost PCB process is $100 \mu\text{m}$. Three metal lines with two spacings are required for a single-channel output to make ground-signal-ground. Therefore, a minimum pitch of $500 \mu\text{m}$ is needed between channels. Considering the beamforming module, each channel is spaced by $500 \mu\text{m}$ in the IC. The antennas are fabricated on a Taconic TLY 5 substrate with a dielectric constant of 2.2. The output of the beamforming IC is connected to antennas with bond-wires, and the dc bias and SPI control are connected to interconnection metals on the FR4 substrate.

A. De-Embedding the Bond-Wire Effect

To reduce the lengths of the bond wires, antennas are fabricated on a TLY 5 substrate with similar heights to the beamforming IC. However, the effect of the bond wires cannot be ignored in the 60 GHz band. Fig. 11 shows the test pattern to model the GSG bond-wires, in which the bond wires connect two 50Ω GCPW ICs. By measuring the 50Ω GCPW IC and the test pattern using a vector network analyzer, the

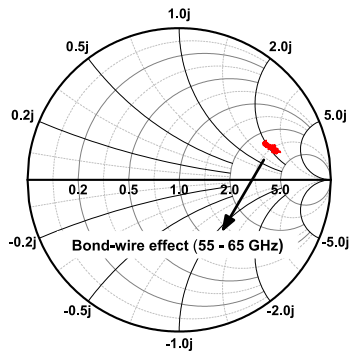


Fig. 12. Impedance of the bond wires in Smith chart.

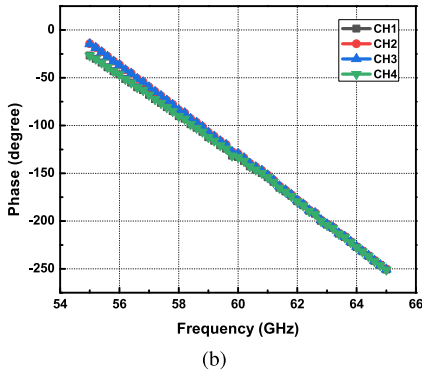
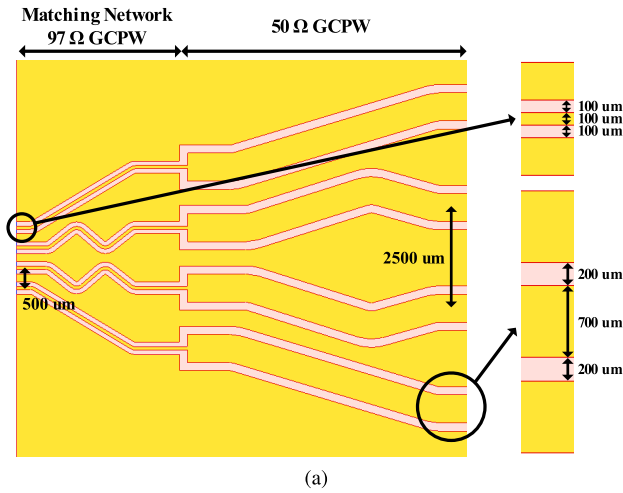


Fig. 13. Antenna feeding network for impedance matching which compensates the effects of the GSG bond-wires. (a) Layout and (b) simulated phases of the feeding network with the GSG bond-wires model.

effect of the bond wires can be extracted. The impedance of the bond wires are presented in Fig. 12.

B. Antenna Feeding Network

The antenna elements' pitch is $2500 \mu\text{m}$ ($\lambda/2$) to reduce the grating lobe, while the spacing between channels in the beamforming IC is $500 \mu\text{m}$. Therefore, the antenna feeding network with impedance matching of the bond wire gradually widens the spacing between channels from 500 to $2500 \mu\text{m}$ while minimizing the phase mismatch between antenna feeding lines.

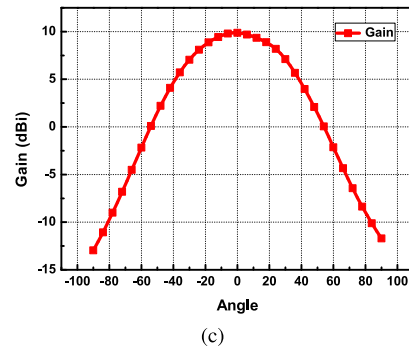
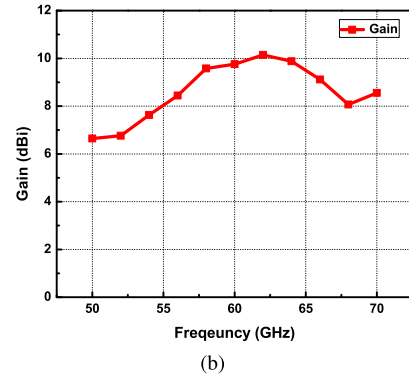
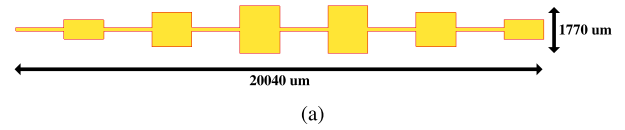


Fig. 14. Six element series-fed patch antenna. (a) Layout, (b) simulated gain with respect to frequency, and (c) simulated gain with respect to angle.

Fig. 13(a) shows the layout of the designed antenna feeding network. The characteristic impedance of the 97Ω GCPW transmission line is utilized to design an impedance matching network. The following 50Ω GCPW transmission line is used to design the antenna feeding network with the spacing between channels of $2500 \mu\text{m}$. Fig. 13(b) shows the phase simulation results for each interconnection consisting of the GSG bond wires, impedance matching network, and 50Ω GCPW transmission line. The phase difference between channels is less than 3.5° at 60 GHz .

C. Six Element Series-Fed Patch Antenna

Fig. 14(a) shows the layout of the six element series-fed patch antenna. Chebyshev weights are applied to reduce the sidelobe level. Fig. 14(b) and (c) show the antenna gain simulation results according to frequency and angle, respectively. The gain of 9.8 dBi is achieved at 60 GHz .

IV. IMPLEMENTATION AND MEASUREMENT RESULTS

Fig. 15 shows the microphotograph of the proposed V-band four-channel transmitter beamforming IC. It is implemented using a 28-nm bulk CMOS process and has a total size of $2200 \times 2000 \mu\text{m}$, including input/output pads. The measured dc power consumption is 295 mW . Fig. 16 shows the measurement setup. The S-parameter and P1dB are measured using an

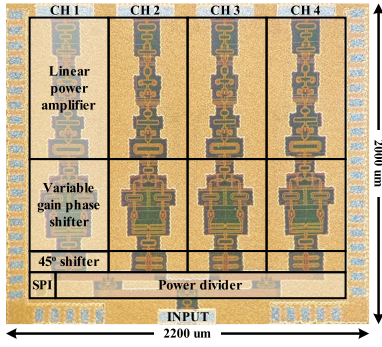


Fig. 15. Microphotograph of the V-band four-channel transmitter beamforming IC.

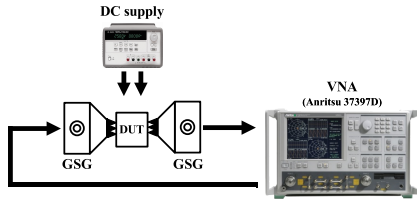


Fig. 16. Measurement setup for S-parameters and P1dB.

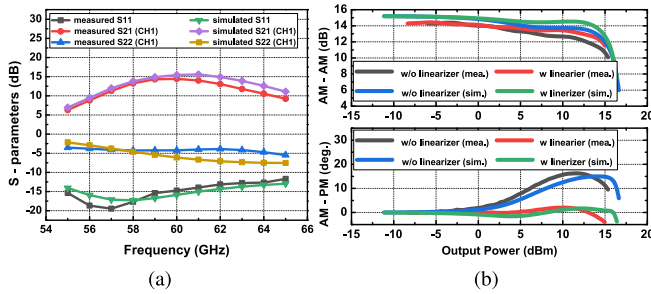


Fig. 17. Measured results of the beamforming IC. (a) S-parameters and (b) gain versus output power (AM to AM) and the phase versus output power (AM to PM).

Anritsu 37397D vector network analyzer. All channels' gain and phase states are controlled by the Total Phase Cheetah SPI host adaptor, and the measurement results are saved through the GPIB cable.

Fig. 17(a) shows the S-parameter measurement results of the beamforming IC. It achieves a peak gain of 14.3 dB in the 3-dB frequency range of 57–63.3 GHz. Fig. 17(b) shows the OP1dB and AM to PM measurement results with the cold-FET-controlled coupled inductor linearizer turned off and on. The proposed linearizer allows the PA to achieve OP1dB of 10.8 dBm and AM to PM performance of less than 2° .

Fig. 18(a) shows the constellation of a channel with 9-bit phase control in 360° and 3-bit gain control in a 20 dB dynamic range at 60 GHz. Fig. 18(b) shows the rms gain and phase errors of the channel. The measured rms gain and phase errors at 60 GHz are 0.1 dB and 2° , respectively, in the total gain states with 3-bit gain control of the 20 dB dynamic range. The measured rms gain and phase errors in the 3-dB frequency bandwidth show 1.06 dB and 3.1° , respectively, in the total gain states with 3-bit gain control of the 20 dB dynamic range. Fig. 19(a) shows the four-channel constellations for

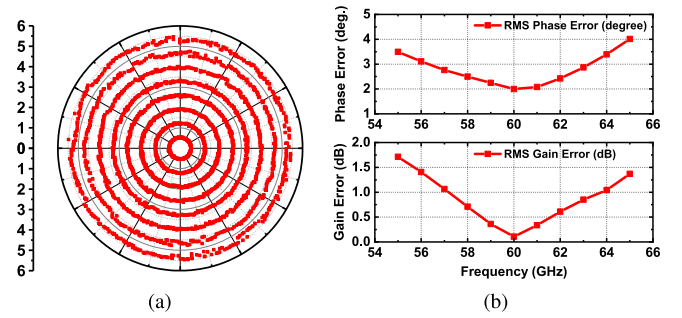


Fig. 18. Measured results of a channel. (a) Constellations for 3-bit gain and 9-bit phase states at 60 GHz and (b) rms gain and phase errors.

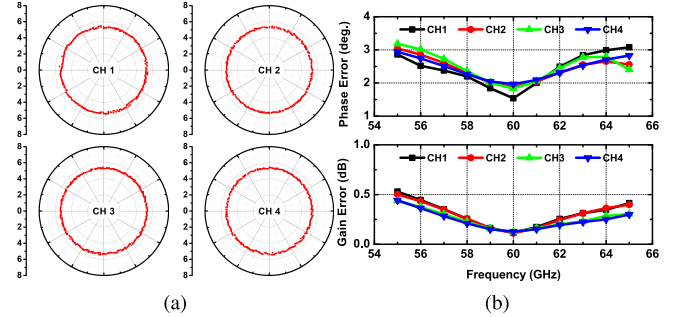


Fig. 19. Measured results. (a) Four-channel constellations for 9-bit phase states at maximum gain state at 60 GHz and (b) rms gain and phase errors.

TABLE I
MEASURED POWER BUDGET

Frequency	60 GHz
OP1dB	10.8 dBm
Bond-wire + antenna feeding network loss	2 dB
Series-fed patch antenna	9.8 dBi
4 channel array gain	12 dB
Path loss	70.9 dB @ 1.4 m
Rx horn antenna gain	20 dBi
Cable loss	8 dB
Received power	-30.9 dBm (measured) -28.3 dBm (calculated)

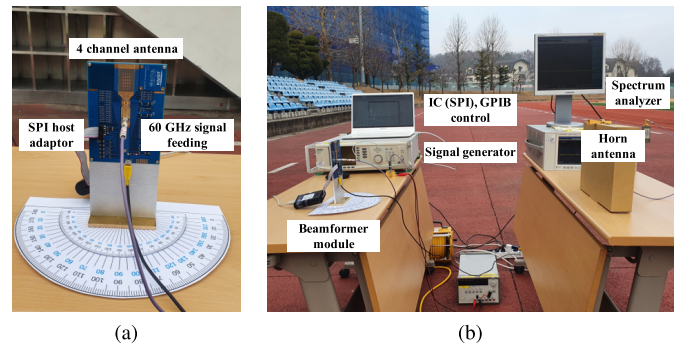


Fig. 20. Antenna measurement setup. (a) Transmitter beamforming antenna module under test and (b) far-field beam pattern measurement setup.

9-bit phase states in maximum gain state at 60 GHz. Fig. 19(b) shows the rms gain and phase errors of the four channels at

TABLE II
PERFORMANCE SUMMARY AND REPORTED V-BAND PHASED-ARRAY TRANSMITTERS

	This work	[21]	[41]	[42]	[43]	[44]
Process	28 nm CMOS	65 nm CMOS	28 nm CMOS	45 nm CMOS	0.18 μ m SiGe	0.18 μ m SiGe
Frequency (GHz)	60	60	60	60	60	60
Number of channel	4	4	8	1	4	64
Gain dynamic range (dB)	20	5.2	9.5	14	18	9
Pdc / channel (mW)	73.8	101	63.5	67	155	95
Phase shifting	Vector Sum + Switched LC	Switched LC	Vector Sum	Switched LC	Vector Sum	Vector Sum
Phase resolution (degree)	0.7	22.5	5	11	11	11
Gain / channel (dB)	20.3	15	24	22	20	18
3-dB bandwidth (GHz)	6.3	11.5	<3*	8	7	6
RMS gain error (dB)	1.06	0.53	0.7	0.7	1.2	1.2
RMS phase error (degree)	2.76	8.8	3.1	5	<9	<9
OP1dB / channel (dBm)	10.8	7.1	10	10	-2	0
Size (mm ²)	4.4	2.88	9.6	0.45	1.8 (Tx + Rx)	N/A

* deduced from graph.

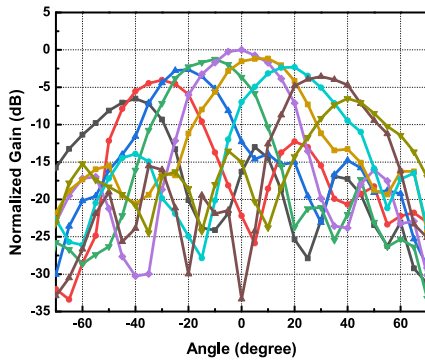


Fig. 21. Measured far-field beam pattern.

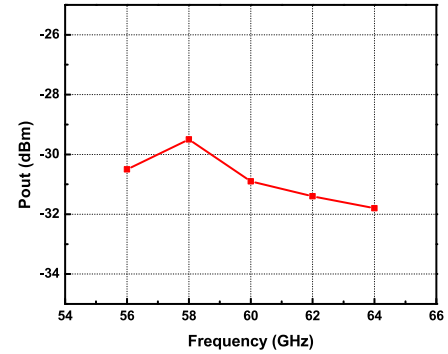


Fig. 22. Measured power of the full-path (spectrum analyzer).

maximum gain state. The measured rms gain and phase errors at 60 GHz are correspondingly 0.12 dB and 2°, respectively. The measured rms gain and phase errors in the 3-dB frequency bandwidth show 0.36 dB and 2.76°, respectively.

Fig. 20 shows the transmitter beamforming module and far-field beam pattern measurement setup. The 60 GHz signal is generated by the Anritsu MG3694C signal generator and Marki MMD-3567L frequency doubler, and it is introduced to the transmitter beamforming IC through a V-band cable and connector. The 60 GHz signal is divided into four channels via power dividers, and each channel determines the gain and phase states. These are radiated through the four channel antennas. DC bias is applied to the IC from the FR4 substrate. The far-field power received with the horn antenna is measured using a Keysight N9030A signal analyzer. A Keysight 11970V harmonic mixer and a Keysight N9029AE13 diplexer are used for the V-band measurement. The phase and gain states in the beamforming IC are controlled through SPI from a laptop. Measurement results were collected and saved from the spectrum analyzer via a GPIB cable. The distance between the transmitter and the horn antenna receiver is 1.4 m, and the beam steering is performed from -40° to 40° .

Fig. 21 shows the results of the far-field beam pattern measurement. It represents the normalized power achieved by setting all channels to the same gain and phase states as a reference. When the beam is steered to normal to antenna plane (0°), the main beam's half power beamwidth (HPBW) achieved about 25° . As the steering angle deviates from 0° to either side, the EIRP of the main lobe decreases due to the beam pattern of the single antenna. Table I represents the power budget when measuring the far-field beam pattern. The measured OP1dB is 10.8 dBm. The total loss of bond wire, impedance matching network, and 50 Ω GCPW transmission line is 2 dB. The gain of the series-fed patch antenna is 9.8 dBi, and the four-channel array gain is 12 dB. At 60 GHz, the path loss at a distance of 1.4 m is 70.9 dB. In the receiver, the horn antenna gain is 20 dBi, and the loss of the V-band cable is 8 dB. Fig. 22 shows the measured power of the full-path. The received power at the signal analyzer is -30.9 dBm, resulting in an additional 2.6 dB loss compared to the calculated results.

Table II summarizes the performance of the proposed four-channel phased array beamformer and compares them with the other state-of-the-art designs.

V. CONCLUSION

A V-band four-channel transmitter beamforming IC using a 28-nm bulk CMOS technology and an antenna module with the IC are presented. A cold-FET-controlled coupled inductor linearizer of a PA is proposed, which allows it to have an OP1dB of 10.8 dBm with a phase variation of less than 2°. In addition, a 45° phase shifter is introduced to detour the vector modulator-based phase shifter's non-ideal effects, resulting in a high phase resolution of 0.7° phase step. The antenna module has four-channel series-fed patch antennas. The chip is connected to the antenna module by GSG wire bonding, of which effects are compensated by impedance matching networks on the substrate. The entire antenna feeding network shows less than 2 dB insertion loss at 60 GHz. The far-field beam pattern of the four-channel transmitter module is measured.

ACKNOWLEDGMENT

The authors would like to thank the Integrated circuit Design Education Center (IDEC) for their support with the CAD tools and Samsung Foundry for their support with a 28-nm bulk CMOS foundry service.

REFERENCES

- [1] C. W. Byeon, C. H. Yoon, and C. S. Park, "A 67-mW 10.7-Gb/s 60-GHz OOK CMOS transceiver for short-range wireless communications," *IEEE Trans. Microw. Theory Techn.*, vol. 61, no. 9, pp. 3391–3401, Sep. 2013.
- [2] R. Wu et al., "64-QAM 60-GHz CMOS transceivers for IEEE 802.11ad/ay," *IEEE J. Solid-State Circuits*, vol. 52, no. 11, pp. 2871–2891, Nov. 2017.
- [3] J. Pang et al., "A 28.16-Gb/s area-efficient 60-GHz CMOS bidirectional transceiver for IEEE 802.11ay," *IEEE Trans. Microw. Theory Techn.*, vol. 68, no. 1, pp. 252–263, Jan. 2020.
- [4] Y. Wang et al., "A 60-GHz 3.0-Gb/s spectrum efficient BPOOK transceiver for low-power short-range wireless in 65-nm CMOS," *IEEE J. Solid-State Circuits*, vol. 54, no. 5, pp. 1363–1374, May 2019.
- [5] E. Öztürk et al., "A 60-GHz SiGe BiCMOS monostatic transceiver for FMCW radar applications," *IEEE Trans. Microw. Theory Techn.*, vol. 65, no. 12, pp. 5309–5323, Dec. 2017.
- [6] A. Kankuppe, S. Park, P. T. Renukaswamy, P. Wambacq, and J. Craninckx, "A wideband 62-mW 60-GHz FMCW radar in 28-nm CMOS," *IEEE Trans. Microw. Theory Techn.*, vol. 69, no. 6, pp. 2921–2935, Jun. 2021.
- [7] W. Lee, T. Dinc, and A. Valdes-Garcia, "Multi-mode 60-GHz radar transmitter SoC in 45-nm SOI CMOS," *IEEE J. Solid-State Circuits*, vol. 55, no. 5, pp. 1187–1198, May 2020.
- [8] H. Forstén, T. Kiuru, M. Hirvonen, M. Varonen, and M. Kaynak, "Scalable 60 GHz FMCW frequency-division multiplexing MIMO radar," *IEEE Trans. Microw. Theory Techn.*, vol. 68, no. 7, pp. 2845–2855, Jul. 2020.
- [9] P. Indirayanti, T. Ayhan, M. Verhelst, W. Dehaene, and P. Reynaert, "A mm-precise 60 GHz transmitter in 40 nm CMOS for discrete-carrier indoor localization," *IEEE J. Solid-State Circuits*, vol. 50, no. 7, pp. 1604–1617, Jul. 2015.
- [10] A. Valdes-Garcia et al., "A fully integrated 16-element phased-array transmitter in SiGe BiCMOS for 60-GHz communications," *IEEE J. Solid-State Circuits*, vol. 45, no. 12, pp. 2757–2773, Dec. 2010.
- [11] A. Natarajan et al., "A fully-integrated 16-element phased-array receiver in SiGe BiCMOS for 60-GHz communications," *IEEE J. Solid-State Circuits*, vol. 46, no. 5, pp. 1059–1075, May 2011.
- [12] M. Tabesh et al., "A 65 nm CMOS 4-element sub-34 mW/element 60 GHz phased-array transceiver," *IEEE J. Solid-State Circuits*, vol. 46, no. 12, pp. 3018–3032, Dec. 2011.
- [13] T. Sowlati, "A 60-GHz 144-element phased-array transceiver for back-haul application," *IEEE J. Solid-State Circuits*, vol. 53, no. 12, pp. 3640–3659, Dec. 2018.
- [14] J. L. Kuo et al., "60-GHz four-element phased-array transmit/receive system-in-package using phase compensation techniques in 65-nm flip-chip CMOS process," *IEEE Trans. Microw. Theory Techn.*, vol. 60, no. 3, pp. 743–756, Mar. 2012.
- [15] E. Cohen, M. Ruberto, M. Cohen, O. Degani, S. Ravid, and D. Ritter, "A CMOS bidirectional 32-element phased-array transceiver at 60 GHz with LTCC antenna," *IEEE Trans. Microw. Theory Techn.*, vol. 61, no. 3, pp. 1359–1375, Mar. 2013.
- [16] I. Nasr et al., "A highly integrated 60 GHz 6-channel transceiver with antenna in package for smart sensing and short-range communications," *IEEE J. Solid-State Circuits*, vol. 51, no. 9, pp. 2066–2076, Sep. 2016.
- [17] B. Sadhu, X. Gu, and A. Valdes-Garcia, "The more (antennas), the merrier: A survey of silicon-based mm-wave phased arrays using multi-IC scaling," *IEEE Microw. Mag.*, vol. 20, no. 12, pp. 32–50, Dec. 2019.
- [18] M. Huang, T. Chi, F. Wang, T. Li, and H. Wang, "A full-FoV autonomous hybrid beamformer array with unknown blockers rejection and signals tracking for low-latency 5G mm-wave links," *IEEE Trans. Microw. Theory Techn.*, vol. 67, no. 7, pp. 2964–2974, Jul. 2019.
- [19] B. Sadhu et al., "A 28-GHz 32-element TRX phased-array IC with concurrent dual-polarized operation and orthogonal phase and gain control for 5G communications," *IEEE J. Solid-State Circuits*, vol. 52, no. 12, pp. 3373–3391, Dec. 2017.
- [20] K. Kibaroglu, M. Sayginer, T. Phelps, and G. M. Rebeiz, "A 64-element 28-GHz phased-array transceiver with 52-dBm EIRP and 8–12-Gb/s 5G link at 300 meters without any calibration," *IEEE Trans. Microw. Theory Techn.*, vol. 66, no. 12, pp. 5796–5811, Dec. 2018.
- [21] J. G. Lee, T. H. Jang, G. H. Park, H. S. Lee, C. W. Byeon, and C. S. Park, "A 60-GHz four-element beam-tapering phased-array transmitter with a phase-compensated VGA in 65-nm CMOS," *IEEE Trans. Microw. Theory Techn.*, vol. 67, no. 7, pp. 2998–3009, Jul. 2019.
- [22] C. So, E.-T. Sung, and S. Hong, "A V-band four-channel phased array transmitter beamforming IC with 0.7-degree phase step in 20 dB dynamic range," in *Proc. IEEE Radio Freq. Integr. Circuits Symp. (RFIC)*, Jun. 2023, pp. 41–44.
- [23] E.-T. Sung, S. Wang, and S. Hong, "A 60-GHz polar vector modulator with lookup table-based calibration," *IEEE Microw. Wireless Compon. Lett.*, vol. 31, no. 6, pp. 572–574, Jun. 2021.
- [24] K.-J. Koh and G. M. Rebeiz, "0.13- μ m CMOS phase shifters for X-, Ku-, and K-band phased arrays," *IEEE J. Solid-State Circuits*, vol. 42, no. 11, pp. 2535–2546, Nov. 2007.
- [25] C. So, E.-T. Sung, and S. Hong, "A 60-GHz variable gain phase shifter based on body floated RF-DAC structure," *IEEE Trans. Circuits Syst. II, Exp. Briefs*, vol. 69, no. 12, pp. 4749–4753, Dec. 2022.
- [26] E.-T. Sung, C. So, and S. Hong, "A 60-GHz variable-gain phase shifter with particular-sized digital-RF cells," *IEEE Trans. Microw. Theory Techn.*, vol. 70, no. 2, pp. 1302–1313, Feb. 2022.
- [27] C. So, E.-T. Sung, and S. Hong, "A 60-GHz variable-gain phase shifter with an active RL poly-phase filter," *IEEE Trans. Microw. Theory Techn.*, vol. 71, no. 2, pp. 593–601, Feb. 2023.
- [28] B.-H. Ku et al., "A 77–81-GHz 16-element phased-array receiver with $\pm 50^\circ$ beam scanning for advanced automotive radars," *IEEE Trans. Microw. Theory Techn.*, vol. 62, no. 11, pp. 2823–2832, Nov. 2014.
- [29] Q. J. Gu, Z. Xu, and M. F. Chang, "Two-way current-combining W-band power amplifier in 65-nm CMOS," *IEEE Trans. Microw. Theory Techn.*, vol. 60, no. 5, pp. 1365–1374, May 2012.
- [30] J. Oh, B. Ku, and S. Hong, "A 77-GHz CMOS power amplifier with a parallel power combiner based on transmission-line transformer," *IEEE Trans. Microw. Theory Techn.*, vol. 61, no. 7, pp. 2662–2669, Jul. 2013.
- [31] B. Park et al., "Highly linear mm-wave CMOS power amplifier," *IEEE Trans. Microw. Theory Techn.*, vol. 64, no. 12, pp. 4535–4544, Dec. 2016.
- [32] D. Kang et al., "A highly efficient and linear class-AB/F power amplifier for multimode operation," *IEEE Trans. Microw. Theory Techn.*, vol. 56, no. 1, pp. 77–87, Jan. 2008.
- [33] S. Jin, B. Park, K. Moon, M. Kwon, and B. Kim, "Linearization of CMOS cascode power amplifiers through adaptive bias control," *IEEE Trans. Microw. Theory Techn.*, vol. 61, no. 12, pp. 4534–4543, Dec. 2013.
- [34] S. Kang, D. Baek, and S. Hong, "A 5-GHz WLAN RF CMOS power amplifier with a parallel-cascoded configuration and an active feed-back linearizer," *IEEE Trans. Microw. Theory Techn.*, vol. 65, no. 9, pp. 3230–3244, Sep. 2017.
- [35] C. So and S. Hong, "A V-band differential push–push frequency doubler with a current-reuse gm-boosted buffer," *IEEE Microw. Wireless Technol. Lett.*, vol. 33, no. 3, pp. 299–302, Mar. 2023.

- [36] K.-Y. Kao, Y.-C. Hsu, K.-W. Chen, and K.-Y. Lin, "Phase-delay cold-FET pre-distortion linearizer for millimeter-wave CMOS power amplifiers," *IEEE Trans. Microw. Theory Techn.*, vol. 61, no. 12, pp. 4505–4519, Dec. 2013.
- [37] S.-M. Weng, Y.-C. Lee, T.-H. Chen, and J. Y. Liu, "A 60-GHz adaptively biased power amplifier with predistortion linearizer in 90-nm CMOS," in *IEEE MTT-S Int. Microw. Symp. Dig.*, Jun. 2018, pp. 651–654.
- [38] T.-S. Chu, J. Roderick, and H. Hashemi, "An integrated ultra-wideband timed array receiver in 0.13 μm CMOS using a path-sharing true time delay architecture," *IEEE J. Solid-State Circuits*, vol. 42, no. 12, pp. 2834–2850, Dec. 2007.
- [39] M.-K. Cho, S.-H. Yoon, S. Sim, L. Jeon, and J.-G. Kim, "CMOS-based bi-directional T/R chipsets for phased array antenna," in *IEEE MTT-S Int. Microw. Symp. Dig.*, Jun. 2012, pp. 1–3.
- [40] S. Lee, J. Park, and S. Hong, "Millimeter-wave multi-band reconfigurable differential power divider for 5G communication," *IEEE Trans. Microw. Theory Techn.*, vol. 70, no. 1, pp. 886–894, Jan. 2022.
- [41] K. Khalaf et al., "A 60-GHz 8-way phased-array front-end with T/R switching and calibration-free beamsteering in 28-nm CMOS," *IEEE J. Solid-State Circuits*, vol. 53, no. 7, pp. 2001–2011, Jul. 2018.
- [42] Y. Wang, H. Chung, Q. Ma, and G. M. Rebeiz, "A 57.5–65.5 GHz phased-array transmit beamformer in 45 nm CMOS SOI with 5 dBm and 6.1% linear PAE for 400 MBaud 64-QAM waveforms," *IEEE Trans. Microw. Theory Techn.*, vol. 69, no. 3, pp. 1772–1779, Mar. 2021.
- [43] B. Rupakula, A. Nafe, S. Zahir, Y. Wang, T.-W. Lin, and G. Rebeiz, "63.5–65.5-GHz transmit/receive phased-array communication link with 0.5–2 Gb/s at 100–800 M and $\pm 50^\circ$ scan angles," *IEEE Trans. Microw. Theory Techn.*, vol. 66, no. 9, pp. 4108–4120, Sep. 2018.
- [44] S. Zahir, O. D. Gurbuz, A. Kar-Roy, S. Raman, and G. M. Rebeiz, "60-GHz 64- and 256-elements wafer-scale phased-array transmitters using full-reticle and subreticle stitching techniques," *IEEE Trans. Microw. Theory Techn.*, vol. 64, no. 12, pp. 4701–4719, Dec. 2016.



Cheol So (Member, IEEE) received the B.S., M.S., and Ph.D. degrees in electrical engineering from Korea Advanced Institute of Science and Technology (KAIST), Daejeon, South Korea, in 2015, 2017, and 2022, respectively.

He is currently a Researcher with the University of California at Santa Barbara, Santa Barbara, CA, USA. His current research interests include millimeter-wave integrated circuits and systems in silicon technologies and multiple-in-multiple-out (MIMO) and phased-array for radars and wireless communications.



radar and high-data-rate wireless communications.

Eun-Taek Sung (Member, IEEE) received the B.S. and M.S. degrees from the School of Electrical Engineering, Chung-Ang University, Seoul, South Korea, in 2014 and 2016, respectively, and the Ph.D. degree in electrical engineering from Korea Advanced Institute of Science and Technology (KAIST), Daejeon, South Korea, in 2021.

He is currently working with Samsung Electronics, Hwaseong, South Korea. His current research interests include millimeter-wave integrated circuits and systems in silicon technologies for high-resolution



Songcheol Hong (Member, IEEE) received the B.S. and M.S. degrees in electronics engineering from Seoul National University, Seoul, South Korea, in 1982 and 1984, respectively, and the Ph.D. degree in electrical engineering and computer science from the University of Michigan, Ann Arbor, MI, USA, in 1989.

He visited the EECS Department, Stanford University, Stanford, CA, USA, as a Visiting Professor in 1997. He was with Samsung Microwave Semiconductor, Milpitas, CA, USA. He served as the Dean of research affairs and the Director of KI-IT Convergence at Korea Advanced Institute of Science and Technology (KAIST), Daejeon, South Korea, where he is currently a Professor with the School of Electrical Engineering and a KT-Chaired Professor. He has authored or coauthored more than 300 technical articles and 150 patents. He is currently interested in RFICs and RF CMOS PAs and especially in millimeter-wave ICs for 5G communications and radars.

Dr. Hong is currently a member of NAEK, KIEES, and KITE. He has served as a Board Member of Techno-Park of Daejeon Metropolitan city. He served as the General Chair for RFIT 2017 supported by the IEEE and TPC Chairs of APMC 2013 and GSMM 2014.